



## Chapter 7

# Debt Securities: Corporate and U.S. Government Loans

### IN THIS CHAPTER

» Understanding the specifics on bonds

» Examining securities issued by the U.S. government

» Looking at short-term bonds

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Instead of giving up a portion of their company (via stock certificates), corporations can borrow money from investors by selling bonds. Local governments (through municipal bonds) and the U.S. government also issue bonds. For SIE exam purposes, most bonds are considered safer than stocks.

Bondholders aren't owners of a company like stockholders are; they're creditors. Bondholders lend money to an institution for a fixed period of time and receive interest for doing so. This arrangement allows the institution to borrow money on its terms (with its chosen maturity date, scheduled interest payments, interest rate, and so on), which it can't do by borrowing from a lending institution.

The SIE exam tests you on your ability to understand the different types of bonds issued, terminology, and yes, some math. This chapter

has you covered in topics relating to corporate and U.S. government debt securities.

## Tackling Bond Terms, Types, and Traits

Before you delve deeper into bonds, make sure you have a good handle on the basics. Understanding the bond basics is a building block that can make all the rest of the bond stuff easier. In this section, I first review basic bond terminology and then move on to some bond characteristics.

### Remembering bond terminology

The SIE exam designers expect you to know general bond terminology. (And I give it to you here — that's why I get paid the big bucks!) In this section, I help you reinforce the information you may have already learned from a prep course or study material (or are in the process of learning). This stuff is basic, but the SIE exam does test it:

- **Maturity date:** All issued bonds have a stated maturity date (for example, 20 years, 30 years, and so on). The maturity date is the year bondholders get paid back for the loans they made. At maturity, bondholders receive par value (see the next bullet).
- **Par value:** Par value is the face value of the bond. Although par value isn't significant to common stockholders (whose issuers use it solely for bookkeeping purposes), it's important to bondholders. For SIE exam purposes, you can assume that the par value for each bond is \$1,000 unless otherwise stated in the question.



**REMEMBER** Bond prices are quoted as a percentage of par value, often without the percent sign. A bond trading at 100 is trading at 100 percent of \$1,000 par. Regardless of whether investors purchase a bond for \$850 (85), \$1,000 (100), or \$1,050 (105), they'll receive par value at the maturity date of the bond, usually with interest pay-

ments along the way. Corporate bonds are usually quoted in increments of 1/8 percent ( $1/8\% = 0.00125$  or \$1.25), so a corporate bond quoted at  $99\frac{3}{8}$  (99.375 percent) would be trading at \$993.75.

- **Coupon rate:** Of course, investors aren't lending money to issuers for nothing; investors receive interest for providing loans to the issuer. The coupon rate on the bond tells the investors how much annual interest they'll receive.

The coupon rate is expressed as a percentage of par value. For example, a bond with a coupon rate of 6 percent would pay annual interest of \$60 ( $6\% \times \$1,000$  par value). You can assume that bonds pay interest semiannually unless otherwise stated. So in this example, the investor would receive \$30 every six months.



**REMEMBER** Bondholders receive *interest* (payment for the use of the money loaned), and stockholders receive *dividends* (see [Chapter 6](#)).

- **The bond indenture:** The *indenture* (also known as *deed of trust*) is the legal agreement between the issuer and the investor. It's printed on or attached to the bond certificate. All indentures contain basic terms:
  - The maturity date
  - The par value
  - The coupon rate (interest rate) and interest payment dates
  - Any collateral securing the bond (see "[Comparing secured and unsecured bonds](#)" later in this chapter)
  - Any callable or convertible features (check out "[Contrasting callable and put bonds](#)" and "[Popping the top on convertible bonds](#)" later in this chapter)

The bond indenture also includes the name of a trustee. A *trustee* is an organization that administers a bond issue for an institution. It ensures that the bond issuer meets all the terms and conditions associated with the borrowing. Essentially, the trustee tries to make sure that the issuer does the right thing.

The following question tests your knowledge of bond interest.



**EXAMPLE** Jane Q. Investor purchased 100 AA rated bonds issued by COW Corp. Jane purchased the bonds at 105 percent of par value, and they are currently trading in the market at 104. If the coupon rate is 7½ percent, how much annual interest does Jane receive?

- (A) \$37.50
- (B) \$75.00
- (C) \$3,750.00
- (D) \$7,500.00

The correct answer is Choice (D). This is a nice, easy question after you wade through the information that you don't need. You need only the number of bonds and the coupon rate to figure out the answer. Don't let yourself get distracted by the rating, purchase price, or market price. That information is there to confuse you.

Jane purchased 100 bonds at \$1,000 par (remember you can assume \$1,000 par) with a coupon rate of 7½ percent, so do the math:

$$100 \text{ bonds} \times \$1,000 \text{ par} \times 7 \frac{1}{2}\% = \$7,500.00$$

Choice (C) would have been correct if the question had asked for the semiannual interest.

## Following bond issue and maturity schedules

Not only can bond certificates be in different forms, but they can also be scheduled with different types of maturities. Maturity schedules depend on the issuer's needs. The following list presents an explanation of the types of bond issues and maturity schedules:

- **Term bonds:** Term bonds are all issued at the same time and have the same maturity date. For example, if a company issues 20 million dollars' worth of term bonds, they may all mature in 20 years. Because of the large payment that's due at maturity, most corporations issuing this type of bond have a sinking fund. Most corporations issue term bonds because they lock in a coupon rate for a long period of time.

A corporation creates a *sinking fund* when it sets aside money over time in order to retire its debt. Investors like to see that a sinking fund is in place because it lowers the likelihood of *default* (the risk that the issuer can't pay interest or par value back at maturity).

- **Series bonds:** These bonds are issued in successive years but have only one maturity date. Issuers of series bonds pay interest only on the bonds that they've issued so far. Construction companies that are building developments in several phases issue this type of bond.
- **Serial bonds:** In this type of bond issue, a portion of the outstanding bonds mature at regular intervals (for example, 10 percent of the entire issue matures yearly). Serial bonds are usually issued by corporations and municipalities to fund projects that provide regular income streams. Most municipal (local government) bonds are issued with serial maturity.

A serial bond that has more bonds maturing on the final maturity date is called a *balloon issue*.



**TIP**

The SIE exam focuses mainly on term and serial bonds. A typical SIE exam question may ask, “Which of the following types of bonds is most likely to have a sinking fund?” Answer: Term bonds.

## Comparing secured and unsecured bonds

The assets of the issuer may or may not back bonds. For test purposes, assume that bonds backed by *collateral* (assets that the issuer owns)

are considered safer for the investor. *Secured bonds*, or bonds backed by collateral, involve a pledge from the issuer that a specific asset (for instance, property) will be sold to pay off the outstanding debt in the event of default. Obviously, with all else being equal, secured bonds normally have a lower yield than unsecured bonds.

The SIE exam tests your knowledge of several types of secured bonds:

- **Mortgage bonds:** These bonds are backed by property that the issuer owns. In the event of default or bankruptcy, the issuer must liquidate the property to pay off the outstanding bonds.
- **Equipment trusts:** This type of bond is mainly issued by transportation companies and is backed by equipment they own (for instance, airplanes or trucks). If the company defaults on its bonds, it sells the assets backing the bonds to satisfy the debt.
- **Collateral trusts:** These bonds are backed by financial assets (stocks and bonds) that the issuer owns. A *trustee* (a financial institution the issuer hires) holds the assets and sells them to pay off the bonds in the event of default.
- **Guaranteed bonds:** Guaranteed bonds are backed by a firm other than the original issuer, usually a parent company. If the issuer defaults, the parent company pays off the bonds. As such, the rating of the bonds is tied to the rating of the guarantor.

*Unsecured bonds* are the opposite of secured bonds: These bonds are not backed by any assets whatsoever, only by the good faith and credit of the issuer. If a reputable company that has been around for a long time issues the bonds, the bonds aren't considered too risky. If they're issued by a relatively new company or one with a bad credit rating, hold onto your seat! Again, for SIE exam purposes, assume that unsecured bonds are riskier than secured bonds. Here's the lineup of unsecured bonds:

- **Debentures:** These bonds are backed only by the issuer's good word and written agreement (the indenture) stating that the issuer

will pay the investor interest when due (usually semiannually) and par value at maturity.

- **Income bonds:** These bonds are the riskiest of all. The issuer promises to pay par value back at maturity and will make interest payments only if earnings are high enough. Companies in the process of reorganization usually issue these bonds at a deep discount (for example, the bonds sell for \$500 and mature at par, or \$1,000). For test purposes (and real-world purposes), you shouldn't recommend these bonds to investors who can't afford to take a lot of risk.



**REMEMBER** Because secured bonds are considered safer than unsecured bonds, secured bonds normally have lower coupon rates. You can assume that for the SIE, the more risk an investor takes, the more reward he will receive. Remember the saying “more risk equals more reward.” More reward may be in the form of a higher coupon rate or a lower purchase price. Either one — or both — lead to a higher yield for the investor.

Check out the following question for an example of how the SIE may test your knowledge of the types of bonds.



**EXAMPLE** Jon Bearishnikoff is a 62-year-old investor who has 50 percent of his portfolio invested in common stock of up-and-coming companies. The other 50 percent of his portfolio is invested in a variety of stocks of more secure companies. Jon would like to start investing in bonds. If Jon's main concern is the safety of his investment, which of the following bonds would you LEAST likely recommend?

**(A)** Collateral trust bonds

**(B)** Mortgage bonds

**(C) Equipment trust bonds**

**(D) Income bonds**

The answer you're looking for is Choice (D). This problem includes a lot of garbage information that you don't need to answer the question. One of your jobs (should you decide to accept it) is to dance your way through the question and cherry-pick the information that you do need. The last sentence is usually the most important one when answering a question. Jon is looking for safety; therefore, you'd least likely recommend income bonds because they're usually issued by companies in the process of reorganizing. As a side note, if you become Jon's broker, he shouldn't have 100 percent of his investments in stock. At his age, Jon should have a decent amount of his portfolio invested in fixed-income securities.



**REMEMBER** When comparing short-term versus long-term debt securities, short-term bonds from the same issuer are considered safer because the investor is not tying up his money for as long a period of time. Because of the extra risk long-term bondholders are taking for tying up their money for a longer period of time, long-term bondholders generally (except in rare cases) receive a higher coupon (interest rate) for taking that additional risk.

## Making Basic Bond Price and Yield Calculations

Although there's not a lot of math on the SIE exam, it does test your knowledge of bond prices, bond yields, and how to calculate them. In this section, I review the relationship between bond prices and bond yields. I also show you how accrued interest can affect how much customers have to pay for the bond. Outstanding bond prices typically don't remain static. As you can imagine, they too are affected by things

like supply and demand, corporate rating change (see “[Determining the Best Investment: Comparing Bonds](#)” later in this chapter), interest rate changes, whether the bond was purchased at a discount (below par value) or a premium (above par value), and so on.



**REMEMBER** The relationship between outstanding bond prices and yields is an inverse one. You can assume for SIE exam purposes that if interest rates decrease, outstanding bond prices increase and vice versa. Say, for example, that a company issues bonds with a 7 percent coupon rate for \$1,000. After the bonds are on the market, interest rates decrease. The company can now issue bonds with a 6 percent coupon rate. Investors with the 7 percent bonds are then in a very good position and can demand a premium for their bonds.

Rates down ↓ = Prices up ↑

Rates up ↑ = Prices down ↓

## Finding bond yields

The following sections review the types of bond yields and how the SIE exam tests this topic.

### Nominal yield (coupon rate)

The *nominal yield* (NY) is the easiest yield to understand because it's the coupon rate on the face of the bonds. For exam purposes, you can assume that the coupon rate will remain fixed for the life of a bond. If you have a 7 percent bond, the bond will pay \$70 per year interest (7% × \$1,000 par value) for the life of the bond. When a problem states that a security is a 7 percent (or 6 percent or whatever) bond, it's giving the nominal yield.

### Current yield

The *current yield* (CY) is the annual rate of return on a security. The CY of a bond changes when the market price changes; you can determine the CY by dividing the annual interest by the market price:

$$\text{Current yield (CY)} = \frac{\text{annual interest}}{\text{market price}}$$

The following question involves bond yields.



**EXAMPLE** Monique Moneybags purchased one XYZ convertible mortgage bond at 105. Two years later, the bond is trading at 98. If the coupon rate of the bond is 6%, what is the current yield of the bond?

- (A) 5.7%
- (B) 6.0%
- (C) 6.1%
- (D) Cannot be determined

The correct answer is Choice (C). Yes, I'm giving you a question with a lot of unnecessary information. All I can tell you is that, unfortunately, you'll have to get used to it. The securities exam creators are notorious for inserting useless (and sometimes misleading) information into the questions to daze and confuse you. In this case, you need only the annual interest and the market price to calculate the answer. Use the following formula to get your answer:

$$\text{CY} = \frac{\text{annual interest}}{\text{market price}} = \frac{\$60}{\$980} = 6.1\%$$

The annual interest is \$60 (6% coupon rate × \$1,000 par value), and the current market price is \$980 (98% of \$1,000 par). The facts that the bond is convertible (bondholders can trade it for common stock — see [“Popping the top on convertible bonds”](#) later in this chapter) or a

mortgage bond (backed by the issuer's property) and that it was purchased at 105 (\$1,050) are irrelevant.



**REMEMBER** “Cannot be determined,” as tempting as it may be, is almost never the correct answer on the SIE exam.

### Yield to maturity (basis)

The *yield to maturity* (YTM) is the yield an investor can expect if holding the bond until maturity. The YTM takes into account not only the market price but also par value, the coupon rate, and the amount of time until maturity. When someone yells to you, “Hey, what’s that bond yielding?” (all right, maybe I run in a different circle of friends), he’s asking for the YTM. The formula for YTM is as follows:

$$\text{YTM} = \frac{\text{annual interest} + \text{annual accretion or} - \text{annual amortization}}{(\text{market price} + \text{par value}) / 2}$$

$$\text{annual accretion} = \frac{\text{par value} - \text{market price}}{\text{years until maturity}}$$

$$\text{annual amortization} = \frac{\text{market price} - \text{par value}}{\text{years until maturity}}$$

I wouldn’t worry about committing the preceding equations to memory. The amount of math you will need on the SIE exam is pretty small. However, it is important that you understand the concepts of measurement and what things like yield to maturity, yield to call, total return, yield to worst, and so on mean.



**TIP**

Accretion and amortization (the adjustment of the bond price toward par over the amount of time until the bond matures), such as in the earlier equation, will not be tested on the SIE exam. However, you will need to be able to calculate them on some of the co-requisite exams. So, the preceding equation should help you visualize how yield-to-maturity works.

## Yield to call

The *yield to call* (YTC) is the amount that the investor receives if the bond is called prior to maturity. The calculations are similar to those for the YTM (see the preceding section), but you substitute the call price for the par value. The chances of needing it on the SIE exam are even more remote than needing the YTM calculations.

## Yield to worst

To determine the *yield to worst* (YTW), you have to calculate the yield to maturity and yield to call for all the call dates (if there's more than one) and choose the lowest. If you get a question on yield to worst, knowing the definition should be enough to get you by.

## Total return

The *total return* calculates the full return on a particular investment over a given period of time. As with the other yields noted, you will likely not need to calculate it but rather to understand it. The total return will provide you with a percentage of gain or loss of an investment. To determine the total return, you need to use the following steps:

1. **Determine the initial cost of the investment.**
2. **Calculate the total amount of interest or dividends received over the time of investment.**
3. **Add the interest or dividends to the selling price.**
4. **Take that number and divide it by the initial cost of the investment and subtract 1.**

## Basis point

A *basis point* is actually one of the easier calculations. Basis points are typically used in the bond market, mutual funds, and exchange-traded

funds (ETFs). Basis points are simply 1/100th of a percent. For example, if the yield on a T-Bond lowers 1.2 percent to 1.1 percent, it is said to have moved ten basis points. Also, when comparing expenses on a fund, people often refer to the fund expenses by basis points. For example, if you are comparing two comparable funds, one fund's expenses might be 0.15 percent and the other's 0.13 percent for a difference of 0.02 percent or two basis points.

## Determining the Best Investment: Comparing Bonds

As you grind your way through SIE exam questions, you may be asked to determine the best investment for a particular investor. You need to carefully look at the question for clues to help you choose the correct answer. (For instance, is the investor looking for safety? Is the investor close to retirement?) Consider several factors, including credit rating, callable and put features, and convertible features.

### Considering bond credit ratings

The institutions that rate bonds are most interested in the likelihood of *default* (the likelihood that the interest and principal won't be paid when due). For the exam, you can assume that the higher the credit rating, the safer the bond and, therefore, the lower the yield.

The two main bond credit rating companies are Moody's and Standard & Poor's (S&P). S&P ratings of BB and lower and Moody's ratings of Ba and lower are considered *junk bonds* or *high-yield bonds*, which have a high likelihood of default, as [Table 7-1](#) explains. (Another credit rating service, called Fitch, uses the same rating symbols as Standard & Poor's.) **Note:** Different sources may show some slight variations in how S&P and Moody's ratings compare; however, the relationships here are the most common.

**TABLE 7-1** Bond Credit Ratings (by Quality)

| <i>Quality</i>                                            | <i>S&amp;P (Standard &amp; Poor's)</i> | <i>Moody's</i> |
|-----------------------------------------------------------|----------------------------------------|----------------|
| Highest                                                   | AAA                                    | Aaa            |
| High                                                      | AA                                     | Aa             |
| Upper medium                                              | A                                      | A              |
| Lower medium                                              | BBB                                    | Baa            |
| Speculative (junk)                                        | BB                                     | Ba             |
| Speculative (junk): Interest or principal payments missed | B                                      | B              |
| Speculative (junk): No interest being paid                | C                                      | Caa            |
| In default                                                | D                                      | D              |

As if these categories aren't enough, S&P can break down each category even further by adding either a plus (+) or minus (–) sign after the letter category. The plus sign represents the high end of the category, and the minus sign designates the lower end of the category. If you see no plus or minus sign, the bond is in the middle of the category.

Moody's can further break down a category by adding a 1, 2, or 3. The number 1 is the highest ranking, 2 represents the middle, and 3 is the lowest. The top four ratings are considered *investment grade*, and the letters below that are considered *junk bonds* or *high-yield bonds*.

**TIP**

The rating company with the capital letters (S&P) uses all capital letters (AAA, AA, and so on). Additionally, S&P has an ampersand (&) between the “S” and the “P” in its name. Think of the ampersand as being like a plus sign to help you remember that S&P uses pluses and minuses within its categories.

Here’s a typical bond-ratings question.

**EXAMPLE**

Place the following Standard & Poor’s bond ratings in order from highest to lowest.

1. A+
2. AA
3. A–
4. BBB+

**(A)** I, II, III, IV

**(B)** I, III, II, IV

**(C)** IV, I, II, III

**(D)** II, I, III, IV

The correct answer is Choice (D). When answering this type of question, always look at the letters first. The only time pluses or minuses come into play is when two answers have the same letters, as in Statements I and III. The highest choice is AA, followed by A+ because it’s higher than A–, which is even higher than BBB+.

## Contrasting callable and put bonds

Your mission for the SIE exam is to know which bonds are better for investors and when bonds are likely to be called or put. As you may know, bonds can be issued in callable and put forms:

- **Callable bonds:** A *callable bond* is a bond that the issuer has the right to buy back from investors at the price stated on the indenture (deed of trust). Callable bonds are riskier for investors because investors can't control how long they can hold onto the bonds. To compensate for this risk, they're usually issued with a higher coupon rate (more risk = more reward).

Most callable bonds are issued with call protection. *Call protection* is the amount of time (usually several years) that an issuer has to wait before calling its bonds. Some callable bonds also have a *call premium*, which is an amount over par value that an issuer has to pay if calling its bonds in the year or years immediately following the expiration of the call protection.

If there is a *make whole call provision*, it allows the issuer to call the bonds providing that the issuer makes a lump sum payment to investors that not only includes payment for the bond but also the present value of any future interest payments investors will miss because of the call.

Another type of bond that can be callable is a *step coupon bond*. Also known as stepped coupon bonds or step-up coupon securities, step coupon bonds typically start at a low coupon rate, but the coupon rate increases at predetermined intervals, such as every five years. The issuer typically has the right to call the bonds at par value at the time the coupon rate is due to increase.

- **Put bonds:** *Put bonds* are better for investors. Put bonds allow the investor to "put" the bonds back (redeem them) to the issuer at any time at the price stated on the indenture. Because the investors have the control, put bonds are (of course) rarely issued. Because these bonds provide more flexibility to investors (who have an interest in the bond and stock prices), put bonds usually have a lower coupon rate.

Remember, there's a direct correlation between interest rates and when bonds are called or put. Issuers call bonds when interest rates decrease; investors put bonds when interest rates increase. Check out the following question to see how this works.



**EXAMPLE** Issuers would call their bonds when interest rates

- (A) increase
- (B) decrease
- (C) stay the same
- (D) are fluctuating

The correct answer is Choice (B). Being adaptable when taking the SIE exam can certainly help your cause. In this question, you have to look from the issuer's point of view, not the investor's. An issuer would call bonds when interest rates decrease because he could then redeem the bonds with the higher coupon payments and issue bonds with lower coupon payments to save money. Conversely, investors would put their bonds back to the issuer when interest rates increase so they could invest their money at a higher interest rate.



**REMEMBER** You can assume for SIE exam purposes that if interest rates increase, bond yields increase.

## Popping the top on convertible bonds

Bonds that are convertible into common stock are called *convertible bonds*. Convertible bonds are attractive to investors because investors

have an interest in the bond price as well as the price of the underlying stock. *Parity* occurs when a convertible bond and its underlying stock (the stock it's convertible into) are trading equally (that is, when a bond trading for \$1,100 is convertible into \$1,100 worth of stock).

**TIP**

If a convertible bond is trading below parity (below the value of the underlying shares it covers), it would make sense for an investor to convert. It costs the investor nothing to convert her bonds into the underlying stock.

In the event that you have to answer a question about convertible bonds to determine whether they're worth converting to the underlying stock, you can use the following formula:

$$\text{conversion ratio} = \frac{\text{par value}}{\text{conversion price}}$$

You can then use the conversion ratio to calculate the parity price:

$$\text{parity price of the bond} = \text{market price of the stock} \times \text{conversion ratio}$$

## Exploring U.S. Government Securities

On the SIE and companion exams, you need to know the basic types of U.S. government securities, their initial maturities, and certain characteristics.

As you may already know, the U.S. government also issues bonds. U.S. government bonds are considered the safest of all securities. Yes, you did read that correctly: the *safest of all securities*. I feel it's worth repeating. How can U.S. government securities be so safe when we're running such a large deficit? Guess what — I don't know, and you don't need to know, either. I can only assume that the U.S. government can always print more currency to make payments on its securities if

needed. However, even U.S. government securities are subject to certain risks such as interest risk, reinvestment risk, purchasing power risk, and so forth (see [Chapter 13](#)).

U.S. government securities are now all issued and held in electronic (book-entry) form. However, because Treasury bonds have maturities of up to 30 years, some are still out there in paper form.



**TIP** With government bonds, you use some of the same types of calculations you use for corporate bonds.

**Understanding the types and characteristics of U.S. government securities**

[Table 7-2](#) gives you an overview of different types of U.S. government securities and their specifics. Memorize all the information in the following chart so you can ace U.S. government securities questions on the SIE exam.

[TABLE 7-2](#) U.S. Government Securities and Time until Maturity

| <i>Security</i>             | <i>Initial Maturity</i>                                                 | <i>Characteristics</i>                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-bills<br>(Treasury bills) | 4, 13, 26, and 52 weeks.<br>Considered short-term U.S. debt securities. | Issued at a discount and mature at par. The difference between the purchase price and par is considered interest even though no interest payments were made. |

| <i>Security</i>                                                                                  | <i>Initial<br/>Maturity</i>                                                                    | <i>Characteristics</i>                                                                                                   |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| T-notes<br>(Treasury<br>notes)                                                                   | 2, 3, 5, 7, and 10<br>years.<br>Considered in-<br>termediate-<br>term U.S. debt<br>securities. | Pay interest every 6<br>months.                                                                                          |
| T-bonds<br>(Treasury<br>bonds)                                                                   | 30 years.<br>Considered<br>long-term U.S.<br>debt securities.                                  | Pay interest every 6<br>months.                                                                                          |
| T-STRIPS<br>(Separate<br>Trading of<br>Registered<br>Interest and<br>Principal of<br>Securities) | 6 months to 30<br>years.                                                                       | Issued at a discount and<br>mature at par.                                                                               |
| TIPS* (Treasury<br>Inflation-<br>Protected<br>Securities)                                        | 5, 10, and 30<br>years.                                                                        | Pay interest every 6<br>months; par value and in-<br>terest payments adjust ac-<br>cording to inflation or<br>deflation. |

*\* TIPS are tied to the Consumer Price Index (CPI), which measures inflation. The par value changes according to inflation. If inflation is high (prices of goods and services are increasing), the par value increases. If we're in a period of deflation (prices on goods and services are decreasing), the par value decreases. Because investors are getting a percentage*

*of par value as their interest payments, the interest payments vary along with the par value.*



**REMEMBER** For the SIE exam, remember that the interest received on U.S. government securities is exempt from state and local taxes. The interest received on municipal bonds is exempt from federal taxes (although I get into that a little more in the next chapter). [Chapter 15](#) gives you the scoop on taxes.

## Agency Securities

Agency bonds are ones issued by a U.S. government-sponsored agency (GSEs — government-sponsored entities). The bonds are backed by the U.S. government, but not all are guaranteed by the full faith and credit of the U.S. government (except for GNMA — Government National Mortgage Association — which are directly backed). As such, agency bonds (although almost as safe) are considered riskier than U.S. government bonds and notes such as T-bonds, T-notes, T-bills, and so on.

GSEs include the following:

- **GNMA** (Government National Mortgage Association — Ginnie Mae): GNMA is the only agency securities backed by the full faith and credit of the U.S. government. GNMA support the Department of Housing and Urban Development (HUD).
- **FNMA** (Federal National Mortgage Association — Fannie Mae): FNMA is a publicly held corporation that is responsible for providing capital for certain mortgages. As such, FNMA may purchase conventional mortgages, VA mortgages, FHA mortgages, and so on. FNMA is privately owned and publicly held but is still government sponsored (a GSE).
- **FHLMC** (Federal Home Loan Mortgage Corporation — Freddie Mac): As a public corporation, Freddie Mac was designed to create

a secondary market for mortgages. Freddie Mac purchases residential mortgages from financial institutions and packages them into mortgage-backed securities that are sold to investors.

- **FCS (Farm Credit System):** The FCS consists of lending institutions that provide financing and credit to farmers. It is a government-sponsored agency but is privately owned. The FCS sells securities to investors and in turn loans the funds raised to farmers. The FCS is overseen by The FCA (Farm Credit Administration).
- **SLMA (Student Loan Marketing Association):** SLMA is not involved in providing mortgages but provides a secondary market for student loans. As such, SLMA purchases student loans and repackages them as short- and medium-term debt securities for sale to investors.



**TIP**

Certain mortgage-backed securities are susceptible to reinvestment risk because many homeowners refinance when interest rates fall (prepayment risk). In that case, holders of mortgage-backed securities get paid back sooner than expected and would be reinvesting at a lower interest rate. In addition, if interest rates stay the same or increase, homeowners will not refinance as often and holders of mortgage-backed securities may end up having to hold their investment for a longer period of time than expected (extension risk).

## Playing It Safe: Short-Term Loans or Money Market Instruments

Your SIE exam will likely include a question or two on money market instruments. *Money market instruments* are relatively safe short-term loans that can be issued by corporations, banks, the U.S. government, and municipalities. Most have maturities of one year or less, and they're usually issued at a discount and mature at par value. The fol-

Following list reviews some basic characteristics of money market instruments to help you earn an easy point or two on the SIE exam:

- **Repurchase agreements:** Repurchase agreements (Repos) are a contract between a buyer and a seller. The seller of the securities (usually T-bills) agrees to buy them back at a previously determined price and time. Repos are short-term loans.
- **Federal Funds:** Federal Funds are loans between banks to help meet reserve requirements. Federal Funds are usually overnight loans for which the rates change constantly depending on supply and demand.



**REMEMBER** *Reserve requirements* are the percentage of deposits that member banks must hold each night. Banks that aren't able to meet their reserve requirements may borrow from other banks at the Fed Funds rate. For more info on the Fed Funds rate and other tools that the Federal Reserve Board uses to influence money supply, see [Chapter 13](#).

- **Corporate commercial paper:** Commercial paper is unsecured corporate debt. Commercial paper is issued at a discount and matures at par value. Commercial paper is issued with an initial maturity of 270 days or less and is exempt from SEC registration.
- **Brokered (negotiable) certificates of deposit:** Brokered CDs are low-risk investments, which originate from a bank and are outsourced to broker-dealers to sell to investors. Unlike typical CDs, which are purchased directly from a bank, brokered CDs can be traded in the market. Negotiable certificates of deposit that require a minimum investment of \$100,000 are often called *jumbo CDs*.
- **Eurodollars:** Eurodollars are American dollars held by a foreign bank outside the U.S. This situation is usually the result of payments made to overseas companies. Eurodollars are not to be confused with Eurodollar bonds (dollar-denominated bonds issued and held overseas).

- **Banker's acceptances:** A banker's acceptance (BA) is a time-draft (short-term credit investment) created by a company whose payment is guaranteed by a bank. Companies use BAs for the importing and exporting of goods.
- **T-bills:** The U.S. government issues T-bills at a discount, and they have initial maturities of 4, 13, 26, or 52 weeks. T-bills are somewhat unique in that they're sold and quoted on a discount-yield basis (YTM). U.S. government securities — and especially T-bills — are considered the safest of all securities.

Here's what a question on money market instruments may look like.



**EXAMPLE** SNK Surfboard Company wants to import boogie boards from an Italian manufacturer in Sicily. SNK would use which of the following money market instruments to finance the importing of the boogie boards?

- (A) T-bills
- (B) Collateral trust bonds
- (C) Repurchase agreements
- (D) Banker's acceptances

The correct answer is Choice (D). You can eliminate Choice (B) right away because collateral trust bonds aren't money market instruments; they're secured long-term bonds. A banker's acceptance is like a post-dated check that's used specifically for importing and exporting goods.

**TIP**

Word association can help you here. If you see *importing*, *exporting*, or *time draft*, your answer is probably banker's acceptance (BA).